BIT503 Fundamentals of Information Technology
Assessment 2

Weighting

40%

Learning outcomes

2 Apply security concepts, tools and techniques to securely manage hardware, software and networking resources required by an organisation and their end users.
3 Apply fundamental networking concepts to a given scenario.
4 Apply the processes and procedures necessary for hardware and software diagnostic testing, maintenance, technical and customer support, and installation and configuration of systems and business applications.

Instructions

Complete and submit your assessment according to the Open Polytechnic's Assessments webpage. This includes information on academic integrity, word limits and referencing.

- Include your name, student number and the assessment number.
- Number your pages.

Submission

- Submit your assessment in one Microsoft Word document.
- Submit your work through your iQualify course.
- Emailed assessments will not be accepted.
- You will receive an automated notice following submission.

By submitting your assessment, you confirm that it is your own, original work.

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Instruction

Read the case study and answer the questions. Your answers, where necessary, should be evidence-based and appropriately cited and referenced.

Case study

It has become very common for fitness centres to have membership programmes. For example, FitLife has the FitRewards programme and HealthZone has the ZonePoints programme. These membership programmes are designed to improve customer satisfaction and increase customer commitment to a brand.

While different membership programmes function slightly differently, there are common features:

- Members are offered discounts on certain classes and services. These discounts tend to vary from week to week.
- Members gain points towards bonuses, such as free classes, personal training sessions or merchandise.
- If members register their accounts, weekly newsletters are sent via email to advertise the weekly specials and class schedules.
- Members who have registered their accounts receive additional emails advertising classes and services based on their activity history.

Aside from the benefits provided to members, fitness centres also use membership programmes to inform their marketing and class offerings.

Another national fitness chain, FitsGo, which has a local branch, is considering introducing their own membership system. They know, though, that the implementation of the above features will require them to put good processes and effective security in place to manage the data they collect.

The CEO of FitsGo has approached the IT management company that you work for to get some advice about how they may do this.

Your manager has asked you to provide the CEO with explanations of the following tasks to aid them in their decision-making.

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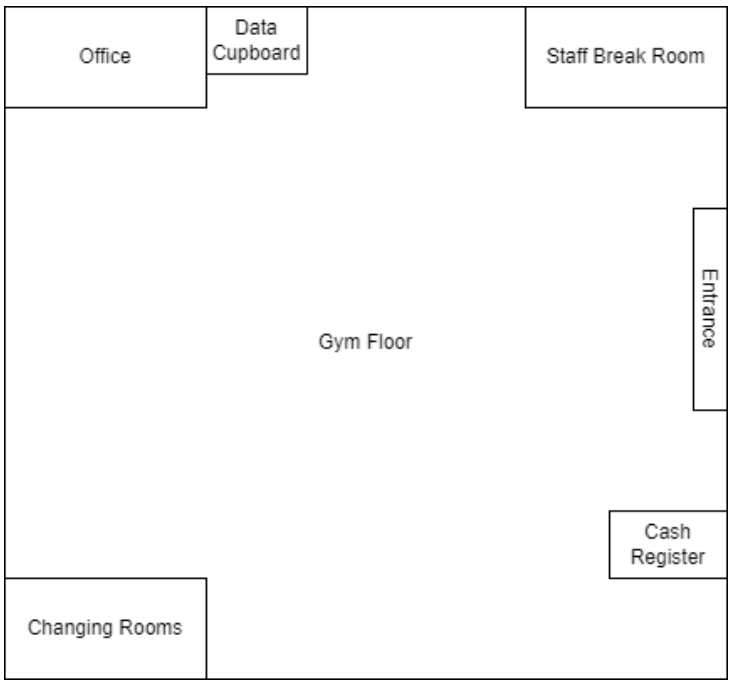
Task 1: Networking

FitsGo is opening a new gym branch in Oamaru. Your manager has asked you to design their network infrastructure. Secure remote access to the head office databases is needed to access business information.

Your manager has provided the following requirements for the network infrastructure:

- The gym branch manager uses a laptop that they would like to use to access the customer database in the gym and at home. They also need to use the printer in the office.
- The office manager works in the designated office in the gym. They work from a wired desktop computer and must be able to print.
- The gym branch and office manager need remote access to the head office databases.
- There is one cash register that has an EFTPOS machine.
- Gym branch staff should have access to Wi-Fi while on breaks.
- The cash register and EFTPOS machine must be connected to the network for internet access and are not wireless.
- All intermediary devices will be securely housed in a locked data cupboard to prevent unauthorised access.

The floor plan is provided for information only.



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Part 1: Network diagram

Use network diagramming software (e.g. Microsoft Visio, LucidChart or similar) to create a logical network diagram for the new gym branch's local area network (LAN).

The network diagram must:

1. contain all network components detailed in the requirements
2. group network components so their physical location is shown using standard network diagram techniques
3. show how the network components connect in a physical topology using wired or wireless connections
4. identify the correct media for connecting devices using a legend
5. provide a connection to the internet protected by a firewall, which provides network services to all devices and gym branch staff.

(20 marks)

Label the diagram according to the following:

- all the physical locations and network devices with their names
- IP addresses to all the wired network devices, including the router
- the gateway address and the device it is assigned to; IP address ranges include:
    - office – 192.168.1.10–192.168.1.19
    - gym floor – 192.168.1.20–192.168.1.29
    - data cupboard – 192.168.1.30–192.168.1.39
- the device that is providing DHCP services.

(10 marks)

Part 2: Network security

FitsGo manages an on-premises data centre at its head office building. Customer data from all branches are stored on a database and backed up hourly to a cloud storage location. Staff at each branch have remote access to the database from desktop computers in their office.

Select two network security tools important for protecting FitsGo's network. Explain how each tool will protect network resources and what each tool protects.

(Word count guideline: 300 words)

(15 marks)

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Part 3: Networking concepts

1. What are the primary responsibilities of a router in a network, and how does it manage traffic between different networks?
2. What features and services does a network switch provide, and how does it enhance the performance of a network for FitsGo?
3. Identify a software solution that ensures secure remote access to a network. Explain its mechanism and its practical application for FitsGo.
4. Describe the process by which devices in a network receive their IP addresses. Discuss the difference between static and dynamic IP addressing and their implications for wired and wireless devices.
5. Compare internet service plans from two different internet service providers (ISPs). Include details of each plan's cost, bandwidth and data limits, and evaluate their suitability for FitsGo.

(Word count guideline: 450 words)

(15 marks)

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Task 2: Diagnostic testing and maintenance

Part 1: Diagnostic tools

The office manager has reported issues with their device but cannot provide specific details about the problem. You only know that the device is not functioning correctly. To identify potential software or hardware issues and replicate the problem, you decide to use diagnostic tools on your personal device.

Choose two diagnostic tools suitable for the device. Run tests using these tools. For each test, provide a screenshot of the results and a comment on the results, focusing on any hardware, software or performance issues identified.

If you are on a Microsoft Windows device, you may choose tools from the following:

- Device Manager
- Resource Monitor
- Task Manager
- Windows Memory Diagnostic.

If you are on an Apple Mac device, you may choose tools from the following:

- Activity Monitor
- Apple Diagnostics
- Disk Utility
- Wireless Diagnostics.

(Word count guideline: 300 words)

(10 marks)

Part 2: Consumption

Use a diagnostic tool to determine the following:

- top two CPU-consuming processes
- top two memory-consuming processes
- top two disk-consuming processes
- top two network-consuming processes.

For each category, provide a screenshot and a comment on the results.

(Word count guideline: 100 words)

(5 marks)

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Part 3: Maintenance

Conduct the following maintenance tasks:

System update

- If you are on a Microsoft Windows device, open Windows Update and take a screenshot showing that your device is up to date.
- If you are on an Apple Mac device, open Software Update and take a screenshot showing that your device is up to date.

Software

- Take a screenshot of the list of software installed on your device.
- Identify any unused software, remove it and take another screenshot showing the updated list.

Disk space

- Take a screenshot showing the amount of free disk space on your device.
- Use appropriate tools to remove unnecessary files (for example, temporary files and browser history).
- Take a screenshot showing the disk space after cleaning.

Backup

- Create a backup of your important files to an external drive or cloud storage.
- Take a screenshot showing the backup process or confirmation of the completed backup.

(15 marks)

Overall presentation and referencing

Marks will be awarded for overall presentation, including grammar, spelling, structure and flow, appropriate acknowledgements, citations and referencing (in APA style).

(10 marks)

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Marking schedule

Task 1 Part 1: Network diagram	16–20 marks	13–15.5 marks	10–12.5 marks	8–9.5 marks	1–7.5 marks
	Thoroughly uses a network diagramming tool to create a logical network diagram for the new gym branch that contains all network components detailed in the requirements. Groups network components to show their physical location using standard network diagram techniques. Clearly shows how the network components connect in a physical topology using wired or wireless connections. Identifies the correct media for connecting devices using a comprehensive legend. Provides a well-detailed connection to the internet protected by a firewall, which offers network services to all devices and gym branch staff.	Clearly uses a network diagramming tool to create a logical network diagram for the new gym branch that contains all network components detailed in the requirements. Groups network components to show their physical location using standard network diagram techniques. Shows how the network components connect in a physical topology using wired or wireless connections. Identifies the correct media for connecting devices using a clear legend. Provides a connection to the internet protected by a firewall, which offers network services to all devices and gym branch staff.	Adequately uses a network diagramming tool to create a logical network diagram for the new gym branch but may lack detail in some areas. Contains most network components detailed in the requirements. Groups network components to show their physical location but may miss some elements. Shows how the network components connect in a physical topology using wired or wireless connections but may not be entirely clear. Identifies the media for connecting devices using a basic legend. Provides a connection to the internet protected by a firewall but may miss some details.	Uses a network diagramming tool to create a logical network diagram for the new gym branch but provides insufficient detail. Contains some network components detailed in the requirements but misses critical components. Groups network components to show their physical location but lacks clarity. Shows how the network components connect in a physical topology but lacks detail or accuracy. Identifies the media for connecting devices, but the legend may be incomplete or unclear. Provides a connection to the internet protected by a firewall but lacks detail.	Provides minimal use of a network diagramming tool to create a logical network diagram for the new gym branch. Fails to contain the network components detailed in the requirements. Does not group network components to show their physical location. Fails to show how the network components connect in a physical topology. Does not identify the media for connecting devices or provide a legend. Does not provide a connection to the internet protected by a firewall.
Task 1 Part 1: Network diagram	**8–10 marks**	**6–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
	Thoroughly labels the network diagram with all physical locations and	Clearly labels the network diagram with all physical locations and	Adequately labels the network diagram with most physical locations	Labels the network diagram with physical locations and network	Provides minimal labelling of the network diagram with physical

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	network devices with their names. Clearly assigns IP addresses to all wired network devices, including the router. Accurately labels the gateway address and the device it is assigned to, using the provided IP address ranges for the office, gym floor and data cupboard. Identifies which device is providing DHCP services.	network devices with their names. Assigns IP addresses to all wired network devices, including the router, with minor errors. Labels the gateway address and the device it is assigned to using the provided IP address ranges for the office, gym floor and data cupboard. Identifies which device is providing DHCP services.	and network devices with their names but lacks detail in some areas. Assigns IP addresses to most wired network devices, including the router, but may miss some devices or have errors. Labels the gateway address and the device it is assigned to but may lack clarity. Uses the provided IP address ranges but may miss some assignments. Identifies which device is providing DHCP services but lacks detail.	devices but provides insufficient detail. Assigns IP addresses to some wired network devices but misses critical devices or has significant errors. Labels the gateway address and the device it is assigned to but lacks clarity and accuracy. Uses the provided IP address ranges but incorrectly. Identifies which device is providing DHCP services but lacks detail and accuracy.	locations and network devices. Fails to assign IP addresses to wired network devices, including the router. Does not label the gateway address or the device it is assigned to. Does not use the provided IP address ranges. Fails to identify which device is providing DHCP services.
Task 1 Part 2: Network security	**12–15 marks**	**10–11.5 marks**	**7.5–9.5 marks**	**6–7 marks**	**1–5.5 marks**
	Thoroughly selects and explains two network security tools important for network protection. Provides detailed explanations of how each tool will protect network resources and what each tool protects, including specific examples and scenarios.	Clearly selects and explains two network security tools important for network protection. Provides clear explanations of how each tool will protect network resources and what each tool protects, with relevant examples.	Adequately selects and explains two network security tools important for network protection but lacks detail in some areas. Provides basic explanations of how each tool will protect network resources and what each tool protects, with some relevant examples.	Selects and explains two network security tools important for network protection but provides insufficient detail. Offers limited explanations of how each tool will protect network resources and what each tool protects, with minimal examples.	Provides minimal selection or explanation of network security tools for network protection. Minimal explanation of how each tool will protect network resources or what each tool protects.
Task 1 Part 3: Networking concepts	**12–15 marks**	**10–11.5 marks**	**7.5–9.5 marks**	**6–7 marks**	**1–5.5 marks**
	Thoroughly explains the primary responsibilities of a router and how it manages traffic and	Clearly explains the primary responsibilities of a router and how it manages traffic.	Adequately explains the primary responsibilities of a router and how it manages traffic.	Briefly explains the responsibilities of a router but provides insufficient detail on	Provides minimal explanation of the primary responsibilities of a router or how it

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	details the features and services of a network switch, including its performance enhancement. Identifies a secure remote access software solution, explaining its mechanism and practical application. Clearly describes how devices receive IP addresses, the difference between static and dynamic IP addressing and their implications for wired and wireless devices. Compares internet service plans from two ISPs, including cost, bandwidth and data limits, and evaluates their suitability comprehensively.	Describes the features and services of a network switch, explaining its performance enhancement. Identifies a secure remote access software solution, explaining its mechanism and practical application. Describes how devices receive IP addresses, the difference between static and dynamic IP addressing and their implications for wired and wireless devices. Compares internet service plans from two ISPs, including cost, bandwidth and data limits, and evaluates their suitability effectively.	Mentions the features and services of a network switch and its performance enhancement. Identifies a secure remote access software solution, explaining its mechanism and practical application in general terms. Describes how devices receive IP addresses, the difference between static and dynamic IP addressing and some implications for wired and wireless devices. Compares internet service plans from two ISPs, including some information on cost, bandwidth and data limits, and provides a basic evaluation of their suitability.	how it manages traffic. Provides limited information on the features and services of a network switch and its performance enhancement. Identifies a secure remote access software solution but provides minimal explanation of its mechanism and practical application. Briefly describes how devices receive IP addresses, mentioning static and dynamic IP addressing with minimal discussion of their implications for wired and wireless devices. Compares internet service plans from two ISPs but provides limited information on cost, bandwidth and data limits and offers a minimal evaluation of their suitability.	manages traffic. Fails to provide sufficient information on the features and services of a network switch or its performance enhancement. Identifies a secure remote access software solution but fails to explain its mechanism or practical application. Provides minimal description of how devices receive IP addresses, with little to no discussion of static and dynamic IP addressing and their implications for wired and wireless devices. Compares internet service plans from two ISPs with insufficient information on cost, bandwidth and data limits and fails to evaluate their suitability.
Task 2 **Part 1: Diagnostic tools**	**8–10 marks**	**6–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
	Selects and runs two diagnostic tools suitable for the device. Provides clear screenshots of the results for each test, with detailed comments on the results. Includes thorough analysis of any hardware, software or	Selects and runs two diagnostic tools suitable for the device. Provides screenshots of the results for each test, with clear comments on the results. Includes a solid analysis of any hardware, software or	Selects and runs two diagnostic tools suitable for the device. Provides screenshots of the results for each test, with adequate comments on the results. Includes a basic analysis of any	Selects and runs two diagnostic tools suitable for the device but provides limited screenshots of the results. Offers minimal comments on the results and a superficial analysis of any hardware,	Provides minimal use of diagnostic tools for the device. Fails to provide screenshots of the results or comments on the results. Offers little to no analysis of any hardware, software or

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	performance issues identified.	performance issues identified.	hardware, software or performance issues identified.	software or performance issues identified.	performance issues identified.
Task 2 Part 2: Consumption	**4–5 marks**	**3.5 marks**	**2.5–3 marks**	**2 marks**	**1–1.5 marks**
	Uses a diagnostic tool to accurately determine the top two CPU-consuming, memory-consuming, disk-consuming and network-consuming processes. Provides clear screenshots for each category and detailed comments on the results.	Uses a diagnostic tool to determine the top two CPU-consuming, memory-consuming, disk-consuming and network-consuming processes. Provides screenshots for each category and clear comments on the results.	Uses a diagnostic tool to identify the top two CPU-consuming, memory-consuming, disk-consuming and network-consuming processes. Provides screenshots for each category and adequately comments on the results.	Uses a diagnostic tool to identify the top two CPU-consuming, memory-consuming, disk-consuming and network-consuming processes but provides limited screenshots. Offers minimal comments on the results.	Provides minimal use of a diagnostic tool to identify the top two CPU-consuming, memory-consuming, disk-consuming and network-consuming processes. Fails to provide sufficient screenshots or comments on the results.
Task 2: Maintenance	**12–15 marks**	**10–11.5 marks**	**7.5–9.5 marks**	**6–7 marks**	**1–5.5 marks**
	Thoroughly conducts all maintenance tasks and provides screenshots for system update, software, disk space and backup.	Clearly conducts all maintenance tasks and provides screenshots for system update, software, disk space and backup.	Adequately conducts all maintenance tasks and provides screenshots for system update, software, disk space and backup.	Conducts some maintenance tasks and provides some screenshots for system update, software, disk space and backup.	Minimal conducting of maintenance tasks and provides minimal screenshots for system update, software, disk space and backup.
Presentation	**8–10 marks**	**6.5–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
Marks will be awarded for overall presentation, including grammar, spelling, structure and flow, appropriate acknowledgements, citations and referencing (in APA style).	Presentation is flawless in all aspects: excellent grammar, logical structure, complete acknowledgements and APA formatting for citations and references, including the reference list without error.	Presentation is with minor issues in grammar or structure, acknowledgements and/or APA formatting for citations and references. Includes a reference list with only minor errors.	Presentation is satisfactory but has several noticeable issues in grammar, structure or APA formatting that moderately affect its quality. Includes a reference list with only minor errors.	Presentation has significant errors in grammar, structure and/or APA formatting. Includes a reference list with some errors.	Presentation is unsatisfactory, with major errors in all aspects, making it unreadable and lacking in academic rigour. Does not include a reference list.

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